



**SCHOOL OF
COMPUTING**

DESIGN AND ANALYSIS OF ALGORITHMS

LAB WORKBOOK

WEEK - 8

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ROLL NUMBER : CH.SC.U4CSE24142

CLASS : CSE-B

Question 1: Write a C program to implement Huffman coding that takes a string as input from the user, calculates the frequency of each character, and generates Huffman codes for all unique characters present in the string. Display the character, its frequency, and the corresponding Huffman code in a tabular format.

Consider the following input string representing a dataset label:

"DATAANALYTICSANDINTELLENCELABORATORY"

CODE:

```
//CH.SC.U4CSE24142 - SANTHOSH A
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#define MAX_TREE_HT 100
#define MAX_CHAR 256
struct MinHeapNode {
    char data;
    unsigned freq;
    struct MinHeapNode *left, *right;
};
struct MinHeap {
    unsigned size;
    unsigned capacity;
    struct MinHeapNode** array;
};
struct MinHeapNode* newNode(char data, unsigned freq) {
    struct MinHeapNode* temp = (struct
MinHeapNode*)malloc(sizeof(struct MinHeapNode));
    temp->left = temp->right = NULL;
    temp->data = data;
    temp->freq = freq;
    return temp;
}
struct MinHeap* createMinHeap(unsigned capacity) {
    struct MinHeap* minHeap = (struct MinHeap*)malloc(sizeof(struct
MinHeap));
    minHeap->size = 0;
    minHeap->capacity = capacity;
    minHeap->array = (struct MinHeapNode**)malloc(minHeap->capacity
* sizeof(struct MinHeapNode));
    return minHeap;
}
void swapMinHeapNode(struct MinHeapNode** a, struct MinHeapNode** b)
{
    struct MinHeapNode* t = *a;
    *a = *b;
    *b = t;
}
```

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void minHeapify(struct MinHeap* minHeap, int idx) {
    int smallest = idx;
    int left = 2 * idx + 1;
    int right = 2 * idx + 2;

    if (left < minHeap->size && minHeap->array[left]->freq <
minHeap->array[smallest]->freq)
        smallest = left;
    if (right < minHeap->size && minHeap->array[right]->freq <
minHeap->array[smallest]->freq)
        smallest = right;
    if (smallest != idx) {
        swapMinHeapNode(&minHeap->array[smallest], &minHeap-
>array[idx]);
        minHeapify(minHeap, smallest);
    }
}

struct MinHeapNode* extractMin(struct MinHeap* minHeap) {
    struct MinHeapNode* temp = minHeap->array[0];
    minHeap->array[0] = minHeap->array[minHeap->size - 1];
    --minHeap->size;
    minHeapify(minHeap, 0);
    return temp;
}

void insertMinHeap(struct MinHeap* minHeap, struct MinHeapNode*
minHeapNode) {
    ++minHeap->size;
    int i = minHeap->size - 1;
    while (i && minHeapNode->freq < minHeap->array[(i - 1) / 2]-
>freq) {
        minHeap->array[i] = minHeap->array[(i - 1) / 2];
        i = (i - 1) / 2;
    }
    minHeap->array[i] = minHeapNode;
}

void buildMinHeap(struct MinHeap* minHeap) {
    int n = minHeap->size - 1;
    for (int i = (n - 1) / 2; i >= 0; --i)
        minHeapify(minHeap, i);
}

int isLeaf(struct MinHeapNode* root) {
    return !(root->left) && !(root->right);
}

void printCodes(struct MinHeapNode* root, int arr[], int top) {
    if (root->left) {
        arr[top] = 0;
        printCodes(root->left, arr, top + 1);
    }
    if (root->right) {

```

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        arr[top] = 1;
        printCodes(root->right, arr, top + 1);
    }
    if (isLeaf(root)) {
        printf(" %c | %3d | ", root->data, root->freq);
        for (int i = 0; i < top; ++i)
            printf("%d", arr[i]);
        printf("\n");
    }
}

void calculateSpaceSaving(struct MinHeapNode* root, int arr[], int
top, int* totalOriginalBits, int* totalCompressedBits) {
    if (root->left) {
        arr[top] = 0;
        calculateSpaceSaving(root->left, arr, top + 1,
totalOriginalBits, totalCompressedBits);
    }
    if (root->right) {
        arr[top] = 1;
        calculateSpaceSaving(root->right, arr, top + 1,
totalOriginalBits, totalCompressedBits);
    }
    if (isLeaf(root)) {
        *totalOriginalBits += root->freq * 8;
        *totalCompressedBits += root->freq * top;
    }
}

void HuffmanCodes(char data[], int freq[], int size) {
    struct MinHeapNode *left, *right, *top;
    struct MinHeap* minHeap = createMinHeap(size);
    for (int i = 0; i < size; ++i)
        minHeap->array[i] = newNode(data[i], freq[i]);
    minHeap->size = size;
    buildMinHeap(minHeap);
    while (minHeap->size != 1) {
        left = extractMin(minHeap);
        right = extractMin(minHeap);
        top = newNode('$', left->freq + right->freq);
        top->left = left;
        top->right = right;
        insertMinHeap(minHeap, top);
    }
    int arr[MAX_TREE_HT], topIdx = 0;
    printf("\nChar | Freq | Huffman Code\n");
    printf("-----\n");
    struct MinHeapNode* root = extractMin(minHeap);
    printCodes(root, arr, topIdx);
    int totalOriginalBits = 0, totalCompressedBits = 0;

```

```

        calculateSpaceSaving(root, arr, 0, &totalOriginalBits,
&totalCompressedBits);
        printf("\nSpace Saving Analysis:\n");
        printf("-----\n");
        printf("Original size (8 bits per character): %d bits\n",
totalOriginalBits);
        printf("Compressed size (Huffman coding): %d bits\n",
totalCompressedBits);
        printf("Space saved: %d bits (%.2f%%)\n",
                totalOriginalBits - totalCompressedBits,
                ((float)(totalOriginalBits - totalCompressedBits) /
totalOriginalBits) * 100);
    }
int main() {
    printf("CH.SC.U4CSE24142 - SANTHOSH A\n");
    char str[1000];
    int freq[MAX_CHAR] = {0};
    printf("Enter the string: ");
    scanf("%s", str);
    for (int i = 0; str[i] != '\0'; i++) {
        freq[(unsigned char)str[i]]++;
    }
    char unique_chars[MAX_CHAR];
    int unique_freqs[MAX_CHAR];
    int n = 0;
    for (int i = 0; i < MAX_CHAR; i++) {
        if (freq[i] > 0) {
            unique_chars[n] = (char)i;
            unique_freqs[n] = freq[i];
            n++;
        }
    }
    HuffmanCodes(unique_chars, unique_freqs, n);
    return 0;
}

```

OUTPUT:

```

D:\AVV CHENNAI\Semester 4\Design and Analysis of Algorithms\Lab Activities\Week 8>gcc Huffman_Coding.c
D:\AVV CHENNAI\Semester 4\Design and Analysis of Algorithms\Lab Activities\Week 8>a
CH.SC.U4CSE24142 - SANTHOSH A
Enter the string: DATAANALYTICSANDINTELLIGENCELABORATORY

Char | Freq | Huffman Code
-----
N | 4 | 000
L | 4 | 001
O | 2 | 0100
R | 2 | 0101
C | 2 | 0110
B | 1 | 01110
G | 1 | 01111
T | 4 | 100
Y | 2 | 1010
S | 1 | 10110
D | 2 | 10111
E | 3 | 1100
I | 3 | 1101
A | 7 | 111

Space Saving Analysis:
-----
Original size (8 bits per character): 304 bits
Compressed size (Huffman coding): 138 bits
Space saved: 166 bits (54.61%)

```

WORKING:

CH-SC-U4C9E2A142

LAB WEEK-8

HUFFMAN CODING

Consider the following input string representing a dataset label:

"DATAANALYTICS AND INTELLIGENCE LABORATORY"

Apply Huffman Coding Algorithm to compress this string -
Also construct a Huffman Tree by showing step by step merging

STEP-1: Frequency Analysis

Letter	Count
A	7
E	4
N	4
T	4
F	3
I	2
C	2
D	2
O	2
R	2
Y	2
B	1
G	1
S	1

STEP-2: Arrange them in Ascending Order:

B(1) - G(1) - S(1) - C(2) - D(2) - O(2) - R(2) - Y(2) - E(3) - T(3) -
N(4) - F(4) - A(7)

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STEP-3: Apply Huffman Algorithm to build the Huffman Tree:-

Initially:

B[1] - G[1] - S[1] - C[2] - D[2] - O[2] - R[2] - Y[2] - E[3] - I[3] - L[4] - N[4] - T[4] - A[7]

Iteration-1: Combine B[1] and G[1] to get BG[2].

S[1] - C[2] - D[2] - O[2] - R[2] - Y[2] - BG[2] - E[3] - I[3] - L[4] - N[4] - T[4] - A[7]

Iteration-2: Combine S[1] and C[2] to get SC[3].

O[2] - O[2] - R[2] - Y[2] - BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4] - A[7]

Iteration-3: Combine D[2] and O[2] to get DO[4].

R[2] - Y[2] - BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4] - DO[4] - A[7]

Iteration-4: Combine R[2] and Y[2] to get RY[4]

BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4] - DO[4] - RY[4] - A[7]

Iteration-5: Combine BG[2] and E[3] to get BGE[5]

BGE[5] - A[7]

I[3] - SC[3] - L[4] - N[4] - T[4] - DO[4] - RY[4] - A[7]

Iteration-6: Combine I[3] and SC[3] to get ISC[6].

L[4] - N[4] - T[4] - DO[4] - RY[4] - BGE[5] - ISC[6] - A[7]

Iteration-7: Combine L[4] and N[4] to get LN[8].

T[4] - DO[4] - RY[4] - BGE[5] - ISC[6] - A[7] - LN[8]

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Iteration - 8: Combine $T[4]$ and $DO[4]$ to get $TDO[8]$

$RY[4] - BGE[5] - ISC[6] - A[7] - LN[8] - TDO[8]$

Iteration - 9: Combine $RY[4]$ and $BGE[5]$ to get $RYBGE[9]$

$ISC[6] - A[7] - LN[8] - TDO[8] - RYBGE[9]$

Iteration - 10: Combine $ISC[6]$ and $A[7]$ to get $ISCA[13]$

$LN[8] - TDO[8] - RYBGE[9] - ISCA[13]$

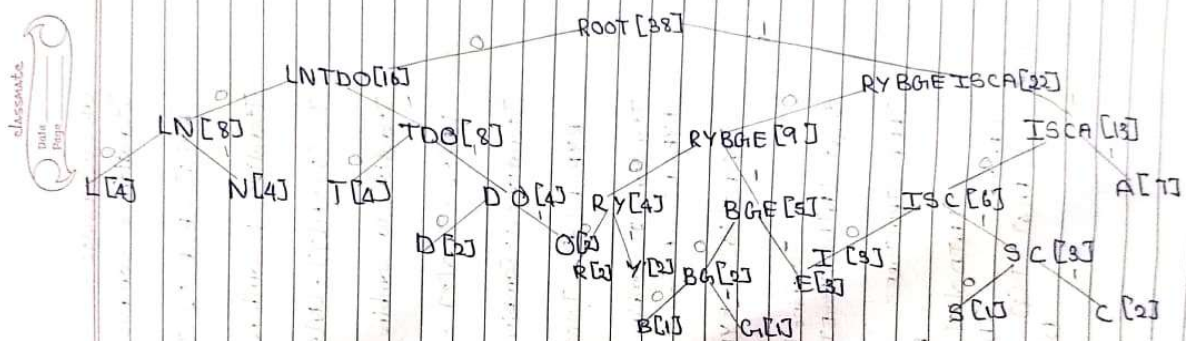
Iteration - 11: Combine $LN[8]$ and $TDO[8]$ to get $LNTDO[16]$

$RYBGE[9] - ISCA[13] - LNTDO[16]$

Iteration - 12: Combine $RYBGE[9]$ and $ISCA[13]$ to get $RYBGEISCA[22]$

$LNTDO[16] - RYBGEISCA[22]$

Combine $LNTDO[16]$ and $RYBGEISCA[22]$ to get $LNTDOORYBGEISCA[38]$



HUFFMAN CODES:

$A = 1111$; $L = 000$; $N = 001$; $T = 010$; $E = 1011$; $I = 1100$; $C = 11011$; $D = 0110$; $O = 0111$

$R = 1000$; $Y = 1001$; $B = 10100$; $G = 10101$; $S = 11010$

Time Complexity: $O(N + M \log M)$: $O(n + m \log m)$ where N is the input string length and M is the number of unique characters - the frequency counting takes $O(N)$ while building the Huffman tree with m nodes takes $O(M \log M)$ due to repeated extractMin and insert operations in the min-heap.

Space Complexity: $O(N + M)$: $O(N + M)$ where N stores the input string and M stores the Huffman tree nodes and related data structures - since M is bounded by 256 (ASCII characters), this effectively becomes $O(N)$ for practical purposes.